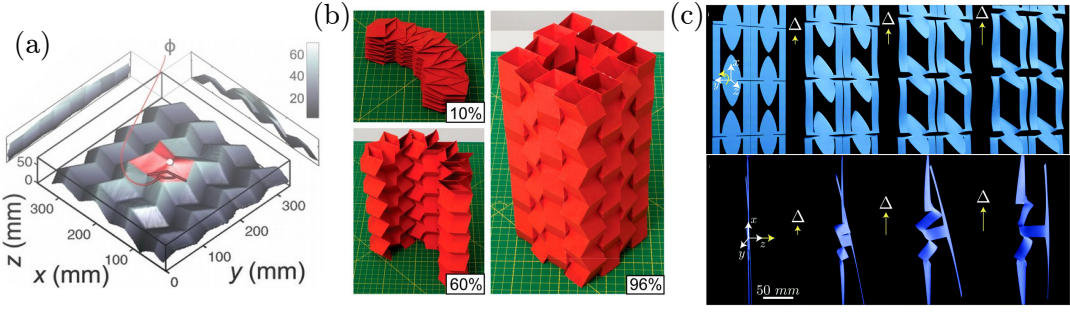

(a)
ϕ
60
40
20
50
0
300
200
100
0
z (mm)
x (mm)
y (mm)
(b)
10%
60%
96%
(c)
Δ
50 mm